Noakhali Science and Technology University
Department of Computer Science and Telecommunication Engineering
4^{th} year 2^{nd} term | Session: 2020-21 | Final Examination: July, 2026
Course Code: CSTE 4203 | Course Title: Machine Learning
Time: 4 hours. | Total Marks: 70

[Answer any seven of the following questions. Figures in the right-hand margin indicate full marks]

1. a) A machine learning model is trained using a small training dataset. Suppose the size of the training dataset is significantly increased while the data quality remains the same. How is the model's performance on unseen data likely to change? Justify your answer. 3

   b) An online shopping company trains two machine learning models for product recommendation. 4
   - Model A is simple and makes a few errors on the training data.
   - Model B is highly complex and correctly classifies every training example.

   Discuss which model is expected to perform better on unseen customer data. Explain your answer using the concepts of over fitting and generalization.

   c) A customer is represented by the following information: 3

Age	Income	Credit Score	Purchased
30	65,000	Good	Yes

   Write the feature vector. If the dataset is linearly separable, what type of decision boundary would you expect the classifier to learn?

2. a) Illustrate why the ROC curve is generated by varying the classification threshold rather than using a single threshold. 3

   b) A data scientist is preparing an imbalanced dataset for binary classification. They are evaluating two different data manipulation techniques before training: Stratified Splitting and Bootstrapping (with replacement). 3

   i. Explain how Stratified Splitting alters or preserves the original distribution of the target classes when creating a test set.

   ii. Describe the fundamental mechanism of Bootstrapping.

   c) A researcher compares the prediction accuracies of two machine learning models using a paired t-test. The test produces the following results: 4

   Significance level $(\alpha) = 0.05$ and p-value $= 0.018$

   i. Interpret the p-value and state the conclusion of the hypothesis test.

   ii. Explain what this result implies about the performance of the two models.

3. a) A music streaming startup wants to predict whether a premium user will **Cancel** their subscription next month based on their behavior over the last 30 days. 6

User ID	Plan Type	Daily Active Minutes	Received Discount	Cancelled? (target)
1	Family	12	No	Yes
2	Individual	45	Yes	No
3	Student	80	No	No
4	Individual	15	No	Yes
5	Student	50	Yes	No
6	Family	85	No	No
7	Student	0	No	Yes
8	Individual	8	Yes	Yes

4

4

0.515

   Which attribute will be chosen by the Information Gain criterion as the decision tree root and why? Show the detail calculation.

   b) A decision tree uses the following splitting conditions and leaf predictions: 4

   - If $X_1 \leq 4$:
     - If $X_2 \leq 3$, predict **Class A**.
     - Otherwise, predict **Class B**.

- If $X_1 > 4$:
    - If $X_1 > 7$, predict Class C.
    - Otherwise, predict Class D.

i. Draw the axis-parallel decision boundaries in the X_1-X_2 plane.
ii. Label each decision region with its predicted class and state the total number of decision regions.

4. a) How does the Naïve Bayes classifier work? Explain. 4

b) You are given the following training data for spam classification. Calculate the posterior probability that a new email with words "Offer=Yes" and "Win=Yes" is Spam, using Naïve Bayes with Laplace smoothing. 4

Email	Offer	Win	Class
1	Yes	Yes	Spam
2	Yes	No	Spam
3	No	Yes	Spam
4	Yes	Yes	Not Spam
5	No	No	Not Spam

c) Can you elaborate the concept of class conditional independence as it applies to the Naïve Bayes classifier? 2

5. a) A dataset contains 20 features, but only a few of them are informative. Explain how Random Forest reduces the influence of irrelevant features during training. 3

b) Illustrate how ensemble learning enhances the predictive performance of perceptron-based classifiers. How does combining multiple perceptrons expand the hypothesis space? 3

c) Two weak classifiers are trained during an AdaBoost run: 4

Classifier A: Weighted error $\epsilon_A = 0.10$
Classifier B: Weighted error $\epsilon_B = 0.25$

Using the standard AdaBoost voting weight formula $\alpha_t = \frac{1}{2} ln \frac{1-\epsilon_t}{\epsilon_t}$, calculate the exact numeric voting weights for both classifiers. Prove which one dominates the final vote.

6. a) A Gradient Boosting Classifier initializes its base model $F_0(x)$ using the log-odds of the positive class. 4

i. Calculate the initial probability (p) of a user cancelling their subscription based purely on the base dataset (Question 3(a)). Then, use the log-odds formula to find the exact numerical value of the initial prediction baseline $F_0(x)$:

$$F_0(x) = \ln(\frac{p}{1-p})$$

ii. Convert this initial log-odds baseline prediction back into a probability value using the logistic sigmoid function:

$$p_0 = \frac{1}{1+e^{-F_0(x)}}$$

Based on this starting probability p_0, calculate the initial pseudo-residual (error) for user 3 (who did not cancel) and user 4 (who did cancel).

b) A leaf node in a Gradient Boosting Classifier contains exactly three users with the following current prediction probabilities (p) and actual churn outcomes (y): 3

- *User A:* y = 1, p = 0.7
- *User B:* y = 1, p = 0.6
- *User C:* y = 0, p = 0.4

Calculate the exact terminal leaf weight for this node using the classification leaf formula. Show your work.

c) Differentiate between AdaBoost and Gradient Boosting with respect to learning strategy, handling of errors and weak learner training. 3

7. a) Explain how sigmoid function work in logistic regression and why it is not a regression model even though it name has it? 3

b) Predict the probability of sudden death for a 50 year old man with systolic blood pressure of 120 mmHg, a relative weight of 100% a cholesterol level of 250 mg/100mL, a glucose level of 100 mg/100mL, a hematocrit of 40%, and a vital capacity of 450 centiliters who smokes 10 cigarettes per day. (Note that these numbers are near average for a healthy man except for the cholesterol level which is high, and of course the number of cigarettes smoked.) 3

c) Does SVM only work with linear data points? 2

d) What is the decision boundary in SVM? 2

8. a) What is Hierarchical Clustering? 2

b) Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters: 5 + 3

A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9).

The distance matrix based on the Euclidean distance is given below:

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	$\sqrt{25}$	$\sqrt{36}$	$\sqrt{13}$	$\sqrt{50}$	$\sqrt{52}$	$\sqrt{65}$	$\sqrt{5}$
A2		0	$\sqrt{37}$	$\sqrt{18}$	$\sqrt{25}$	$\sqrt{17}$	$\sqrt{10}$	$\sqrt{20}$
A3			0	$\sqrt{25}$	$\sqrt{2}$	$\sqrt{2}$	$\sqrt{53}$	$\sqrt{41}$
A4				0	$\sqrt{13}$	$\sqrt{17}$	$\sqrt{52}$	$\sqrt{2}$
A5					0	$\sqrt{2}$	$\sqrt{45}$	$\sqrt{25}$
A6						0	$\sqrt{29}$	$\sqrt{29}$
A7							0	$\sqrt{58}$
A8								0

Suppose that the initial seeds (centers of each cluster) are A1, A4 and A7. Perform the k-means algorithm for 1 epoch only. At the end of this epoch show:

(i) The new clusters (i.e. the examples belonging to each cluster).

(ii) The centers of the new clusters.

9. a) An input volume of size 64 × 64 × 3 (Width × Height × Channels) is passed through a convolutional layer with 16 filters, a kernel size of 3 × 3, a stride of 1, and same padding (padding = 1). 4

i. Calculate the spatial dimensions (Width × Height × Channels) of the resulting feature map.

ii. Calculate the total number of trainable parameters (including biases) in this convolutional layer.

b) A CNN is trained to classify handwritten digits. The model performs well on the training images but poorly on unseen images. Identify the likely problem and suggest two methods to improve the model. 3

c) Given the following feature map: 3

$$\begin{bmatrix} 2 & 4 & 1 & 3 \\ 5 & 6 & 2 & 1 \\ 3 & 1 & 8 & 4 \\ 2 & 5 & 7 & 6 \end{bmatrix}$$

Apply 2×2 max pooling with stride 2.